

Thrust back on — the trade-off was an artefact

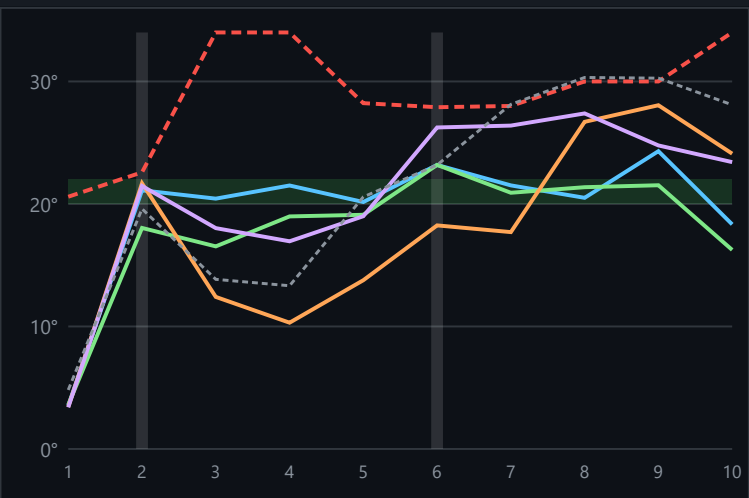
With joint 7 raised to 28° and foot thrust restored, evenness, hip equality and speed improve *together*. They only fought each other on a body that was barely moving.

One shape, thrust turned up



Same wave shape throughout — only the push changes. Speed climbs, bend spread **falls**, hip mismatch **falls**. Past 4 the front joint starts to clip, which sets the ceiling.

How much each joint bends, nose → tail



Red dashed = what the model physically allows (joint 7 now 28°). Green band = the 20–22° target. Pale bars = the two hips. Grey dashed = the old baseline, which over-bends three joints. **The blue line sits inside the band across the whole spine.**

The finalists

EVEN g4 (evenest spine)

push 0.39 · muscle 14 · 1/1/0.7/0.5/0.3 · thrust 4

nothing at its cap

bend **18.3–24.3°**, spread **6.0°**, mean 21.2°

hips **21.1° → 23.2°** (ratio 1.10)

speed **2.85 u/s** wave size 1.58

curvature **0.53%** (heading -5.1°)

roll 7.0/s at peak 1.40°

EVEN g2 (same shape, slower)

push 0.39 · muscle 14 · 1/1/0.7/0.5/0.3 · thrust 2

nothing at its cap

bend **16.2–23.2°**, spread **6.9°**, mean 19.5°

hips **18.0° → 23.2°** (ratio 1.29)

speed **1.61 u/s** wave size 1.27

curvature **0.35%** (heading -2.6°)

roll 8.4/s at peak 1.15°

HIPS g4 (girdle-targeted)

push 0.5 · muscle 14 · 0.8/1/1/0.5/0.5 · thrust 4

nothing at its cap

bend **10.3–28.1°**, spread **17.8°**, mean 19.2°

hips **21.7° → 18.3°** (ratio 0.84)

speed **3.04 u/s** wave size 1.52

curvature **0.51%** (heading -3.9°)

roll 6.5/s at peak 0.81°

AMP g4 (amplitude-targeted)

push 0.5 · muscle 14 · 1/1/0.85/0.5/0.3 · thrust 4

nothing at its cap

bend **17.0–27.4°**, spread **10.4°**, mean 22.6°

hips **21.5° → 26.2°** (ratio 1.22)

speed **2.96 u/s** wave size 1.38

curvature **0.49%** (heading -4.2°)

roll 6.9/s at peak 1.11°

BASE g0 (5.1b baseline, no thrust)

push 0.39 · muscle 18 · 1/1/1/0.6/0.6 · thrust 0

CLIPS seg7, seg8, seg9

bend **13.3–30.3°**, spread **17.0°**, mean 23.0°

hips **19.6° → 23.2°** (ratio 1.18)

speed **0.90 u/s** wave size 1.97

curvature **0.82%** (heading -0.7°)

roll 6.1/s at peak 0.80°

Straightness is fine. The 7–8% sideways figure was the body swimming straight but a few degrees off the reference axis. Measured as actual bend in the path, all four run 0.35–0.53% against the baseline's 0.82% — straighter than what we had.

The one thing that got worse is roll: peak 1.40° against the baseline's 0.80°. Small, real, and worth watching rather than acting on yet.